

SMART INDIA HACKATHON 2026



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HACKATHON
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Problem Statement ID – SIH26102

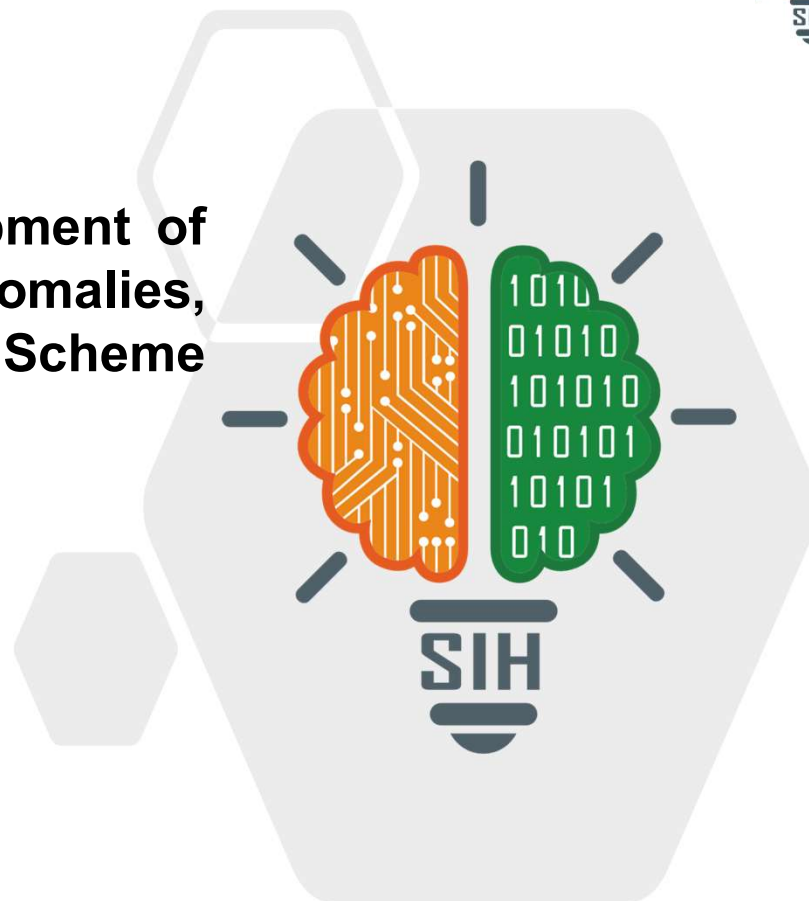
Problem Statement Title - Development of an AI-powered system to detect anomalies, fraud, and inefficiencies in MPLAD Scheme implementation regd.

Theme - Smart Automation

PS Category - Software

Team ID - 124653

Team Name - Civic Sentinel



Civic Sentinel: AI-Powered Risk Engine for MPLADS Monitoring



Proposed Solution: MPLADS Sentinel is an AI intelligence layer over eSAKSHI that detects anomalies, predicts risks, and prioritizes projects for human verification.

How it addresses the problem:

- **Financial Anomalies** — Cost & expenditure deviations.
- **Suspicious Patterns** — Payment, progress & duplicate works.
- **Risk Prediction** — Delay, cost overrun & inefficiency.
- **Explainable Alerts** — Risk-based alerts & escalation.
- **AI Bill Parsing** — Automatically extracts and compares contractor bills against AI estimates to prevent overbilling.
- **Citizen Feedback Loop** — Public petition portal that uses AI to prioritize community needs for authorities.

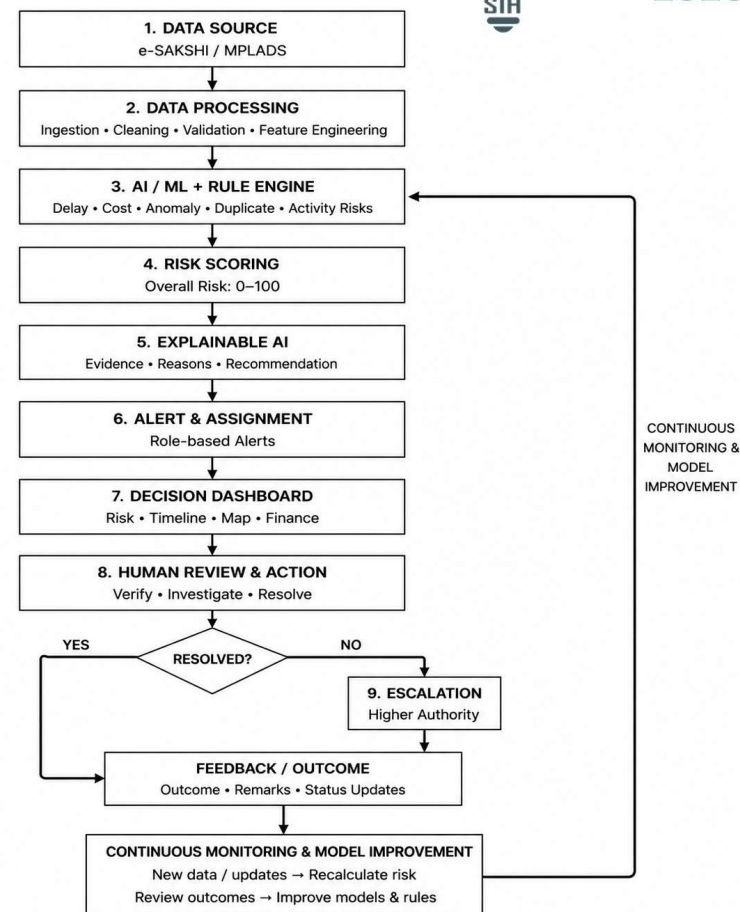
Innovation and uniqueness:

Context-aware AI that transforms eSAKSHI data into predictive risk intelligence, featuring **automated AI bill parsing** and **live citizen-driven project prioritization**.

TECHNICAL APPROACH



Category	Technology / Tools
Frontend / Dashboard	React.js, Tailwind CSS, Leaflet
Backend / APIs	Python, FastAPI, REST APIs
Database	PostgreSQL, PostGIS, pgvector
AI / ML	Python, Scikit-learn, XGBoost, Sentence Transformers
RAG / Documents	LlamaIndex, pgvector, PyMuPDF, OCR
Data Processing	Pandas, NumPy
Deployment	Docker, AWS / GCP



FEASIBILITY AND VIABILITY



Technical Feasibility –
Uses proven AI/ML, RAG, PostgreSQL, and geospatial technologies to analyze MPLAD project data.

Economic Feasibility –
A scalable software solution that minimizes infrastructure costs and reduces manual inspection workload.

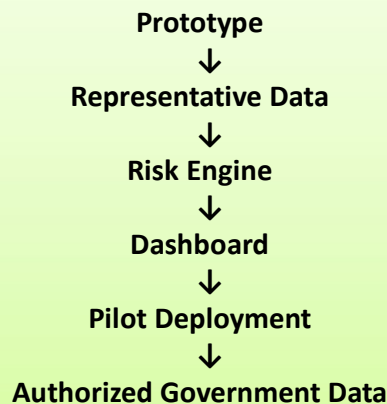
Operational Feasibility
– Provides officials with a simple dashboard for risk scores, evidence, alerts, and investigation priorities, with role-based escalation for unresolved cases.

Governance Feasibility
– Strengthens transparency and accountability through early detection of irregularities in public-funded projects.

Challenges & Resolution:

- **Data Access:** Use authorized APIs/data exchange or government-provided datasets; prototype with publicly available or representative data.
- **False Positives:** Use context-aware benchmarks, multiple risk signals and human verification instead of automatically declaring fraud.
- **Data Security & Privacy:** Role-based access, encryption and secure handling of government/project data.

Implementation Strategy



IMPACT AND BENEFITS



IMPACT

- ❖ Greater Transparency
- ❖ Faster Risk Identification
- ❖ Improved Fund Monitoring
- ❖ Stronger Public Accountability

BENEFITS

- ❖ Reduced Manual Inspection
- ❖ Early Detection of Irregularities
- ❖ Efficient Resource Allocation
- ❖ Evidence-Based Decision Making

What-if Simulation:

Estimate how intervention could affect project delay and cost risk.

Continuous Learning:

Improve risk detection using verified investigation outcomes.

CORE CAPABILITIES

Advanced Document AI:

Automated invoice, estimate and document verification.

Geospatial Intelligence:

Uses GPS coordinates and available map imagery for project location verification and visual assessment.



- **MPLADS Guidelines, 2023:** Official guidelines governing recommendation, sanction, execution, monitoring and completion of MPLADS works. [MPLADS Guidelines 2023](#)
- **eSAKSHI MPLADS Portal:** Official source for MPLADS work and expenditure data used as the basis for project analysis. [MPLADS](#)
- **CAG Audit Reports (Performance & Compliance):** Documented findings of irregular payments, monitoring weaknesses, and real-world fraud patterns. [CAG Performance Audit](#) and [CAG Compliance Audit](#)
- **MoSPI Annual Report 2023–24:** Official overview of MPLADS structure, implementation responsibilities and current guidelines. [MoSPI Annual Report 2023–24](#)
- **Live Prototype:** <https://civic-sentinel-ai.vercel.app/>
- **Project Report:** https://drive.google.com/file/d/1XDdleYGR_8vknqZVJ7fMvAGPxmnHraeG/view?usp=sharing
- **Video Demo:** https://youtu.be/q9lyEGdKg_c